ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

$$f_x = 3200 pixels; field of view = 160 mm; 1 pixel = 0.125 mm$$

Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 2 [10 marks]

Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\, x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since $|\sin \theta_e|$ less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to $mg\ell$ = 2.4525 N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \leq \sin\theta_e \leq 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

With $x = [\theta, \dot{\theta}]^T$, A = [[0,1], [-(mgℓ/J)cos θ_e, -b/J]], B = [[0],[1/J]]. At θ_e = 60°, A = [[0,1], [-61.3125, -2.5]] and B = [[0],[50]].

At the equilibrium, the constant torque balances gravity. For a small perturbation, use $\sin(\theta_e + \delta\theta) \approx \sin\theta_e + \cos\theta_e\, \delta\theta$. The constant terms cancel against τ_0; the remaining linear term gives the local state matrix.

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is Kp(θd-θ), which must balance mgℓ sin θ.

Q2(d) [2 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

For H(s)=ωc/(s+ωc), ωc=2π(20)=125.664 rad/s and T=0.001 s. Tustin gives H(z)=α(1+z^{-1})/[(1+α)+(α-1)z^{-1}], α=ωcT/2=0.0628319. Thus:

$$y[k] = 0.881765y[k-1] + 0.0591175(x_k + x[k-1])$$

DC gain = (2x0.0591175)/(1-0.881765) approximately 1.000.

$\omega_c = 2\pi(20) = 125.664$ rad/s. With $T = 0.001$ s, $\alpha = \omega_c T/2 = 125.664(0.001)/2 = 0.062832$. The recurrence coefficients are approximately 0.881765 on the previous output and 0.0591175 on each input sample. Check: $2(0.0591175)/(1 - 0.881765) \approx 1$, so the DC gain is close to one.

Q3(c) [4 marks]

$$L = [18, 64]^T$$

Q3(d) [2 marks]

An observer uses the plant model and known control input to estimate unmeasured states such as velocity, and its poles can be designed. A simple low-pass filter cannot exploit the state-space model in this way. A filter may be safer when the model is poorly known or unreliable, because it relies less on model accuracy.

The observer uses the plant model and the measured output to reconstruct the unmeasured state. Its convergence depends on the model and the observer poles. A low-pass filter is simpler, but generally introduces phase lag and does not explicitly use the plant dynamics.

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) **[4 marks]**

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) **[2 marks]**

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.